PHB_221_Final_Project

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2025-11-13

Main Questions

Main Question: How does the age-adjusted death rate for heart disease vary across states in the United States, and are there any observable regional or geographic trends over time?

- Research Objectives:
  - Compare state-level heart disease mortality rates
  - Identify patterns
  - Explore how rates change over time
  - Determine whether areas with consistently higher mortality are geographically concentrated
  - Can past trends in age adjusted heart disease death rates be used to predict future state level rates?

Dataset and Libraries

```r
# libraries
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3

## Warning: package 'purrr' was built under R version 4.3.3

## Warning: package 'lubridate' was built under R version 4.3.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.3.3

## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
```

```
##     expand, pack, unpack
```

```r
library(Metrics)
```

```
## Warning: package 'Metrics' was built under R version 4.3.3
```

```r
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.3.3

## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```r
library(alr4)
```

```
## Loading required package: car

## Warning: package 'car' was built under R version 4.3.3

## Loading required package: carData

## Warning: package 'carData' was built under R version 4.3.3

##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
##
## Loading required package: effects
## lattice theme set by effectsTheme()
## See ?effectsTheme for details.
```

```r
# dataset
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv")
```

```
## Rows: 10868 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): 113 Cause Name, Cause Name, State
## dbl (3): Year, Deaths, Age-adjusted Death Rate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Data Collection and Cleaning

```r
# view first few rows of data
head(data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                   <dbl>
## 1  2017 Accidents (unintention~ Unintention~ Unit~ 169936                    49.4
## 2  2017 Accidents (unintention~ Unintention~ Alab~   2703                    53.8
## 3  2017 Accidents (unintention~ Unintention~ Alas~    436                    63.7
## 4  2017 Accidents (unintention~ Unintention~ Ariz~   4184                    56.2
## 5  2017 Accidents (unintention~ Unintention~ Arka~   1625                    51.8
## 6  2017 Accidents (unintention~ Unintention~ Cali~  13840                    33.2
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
# check the number of rows and cols
dim(data)
```

```
## [1] 10868     6
```

```r
# filter data
filtered_data <- data[data$`Cause Name` == "Heart disease", ]

head(filtered_data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                   <dbl>
## 1  2017 Diseases of heart (I00~ Heart disea~ Unit~ 647457                     165
## 2  2017 Diseases of heart (I00~ Heart disea~ Alab~  13110                     223.
## 3  2017 Diseases of heart (I00~ Heart disea~ Alas~    814                     135
## 4  2017 Diseases of heart (I00~ Heart disea~ Ariz~  12398                     142.
## 5  2017 Diseases of heart (I00~ Heart disea~ Arka~   8270                     224.
## 6  2017 Diseases of heart (I00~ Heart disea~ Cali~  62797                     143.
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
dim(filtered_data)
```

```
## [1] 988   6
```

```r
# missing data
colSums(is.na(filtered_data))
```

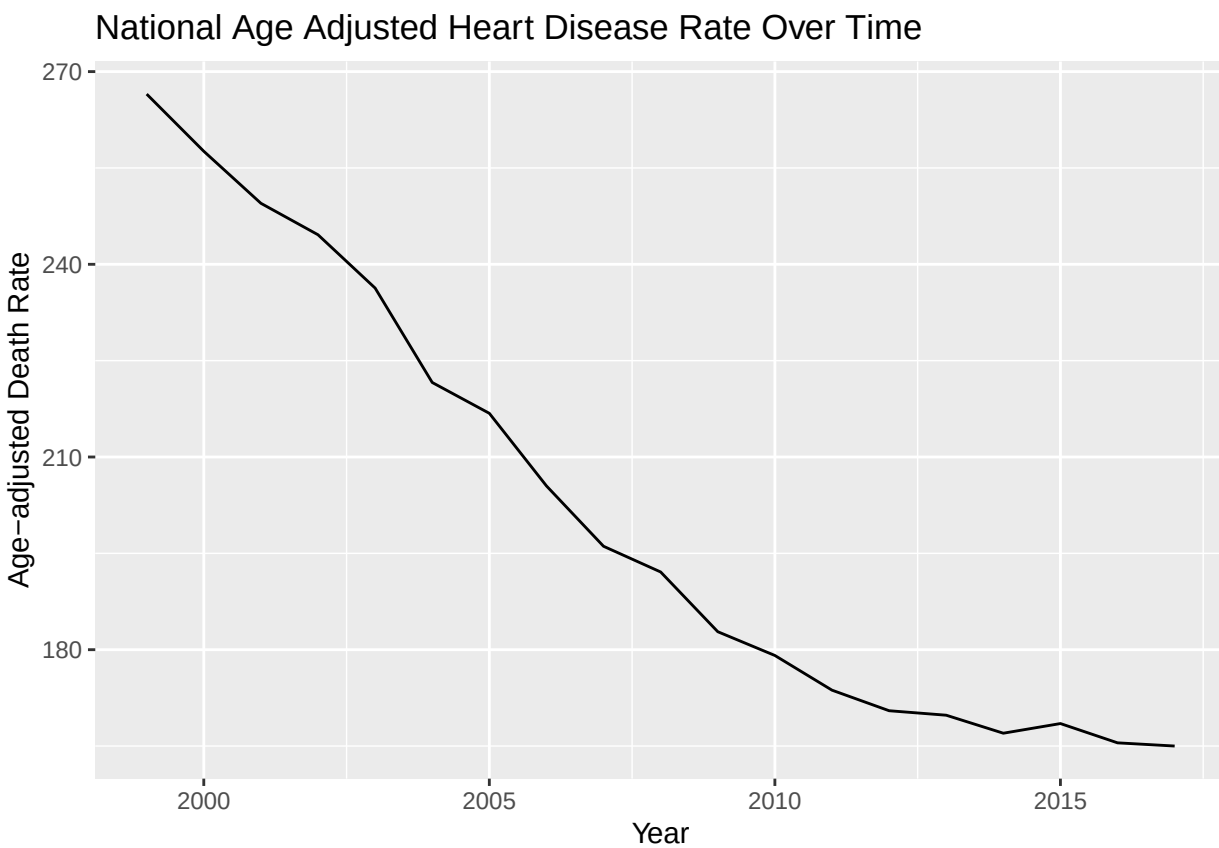
```
##                     Year          113 Cause Name              Cause Name
##                        0                       0                       0
##                    State                  Deaths Age-adjusted Death Rate
##                        0                       0                       0
```

Explantory Analysis

```r
# national level
national_data <- data |>
  filter(`Cause Name` == "Heart disease", State == "United States")

head(national_data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                   <dbl>
## 1  2017 Diseases of heart (I00~ Heart disea~ Unit~ 647457                    165
## 2  2016 Diseases of heart (I00~ Heart disea~ Unit~ 635260                    166.
## 3  2015 Diseases of heart (I00~ Heart disea~ Unit~ 633842                    168.
## 4  2014 Diseases of heart (I00~ Heart disea~ Unit~ 614348                    167
## 5  2013 Diseases of heart (I00~ Heart disea~ Unit~ 611105                    170.
## 6  2012 Diseases of heart (I00~ Heart disea~ Unit~ 599711                    170.
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
ggplot(data = national_data, mapping = aes(x = Year, y = `Age-adjusted Death Rate`)) +
  geom_line() +
  labs(title = "National Age Adjusted Heart Disease Rate Over Time")
```

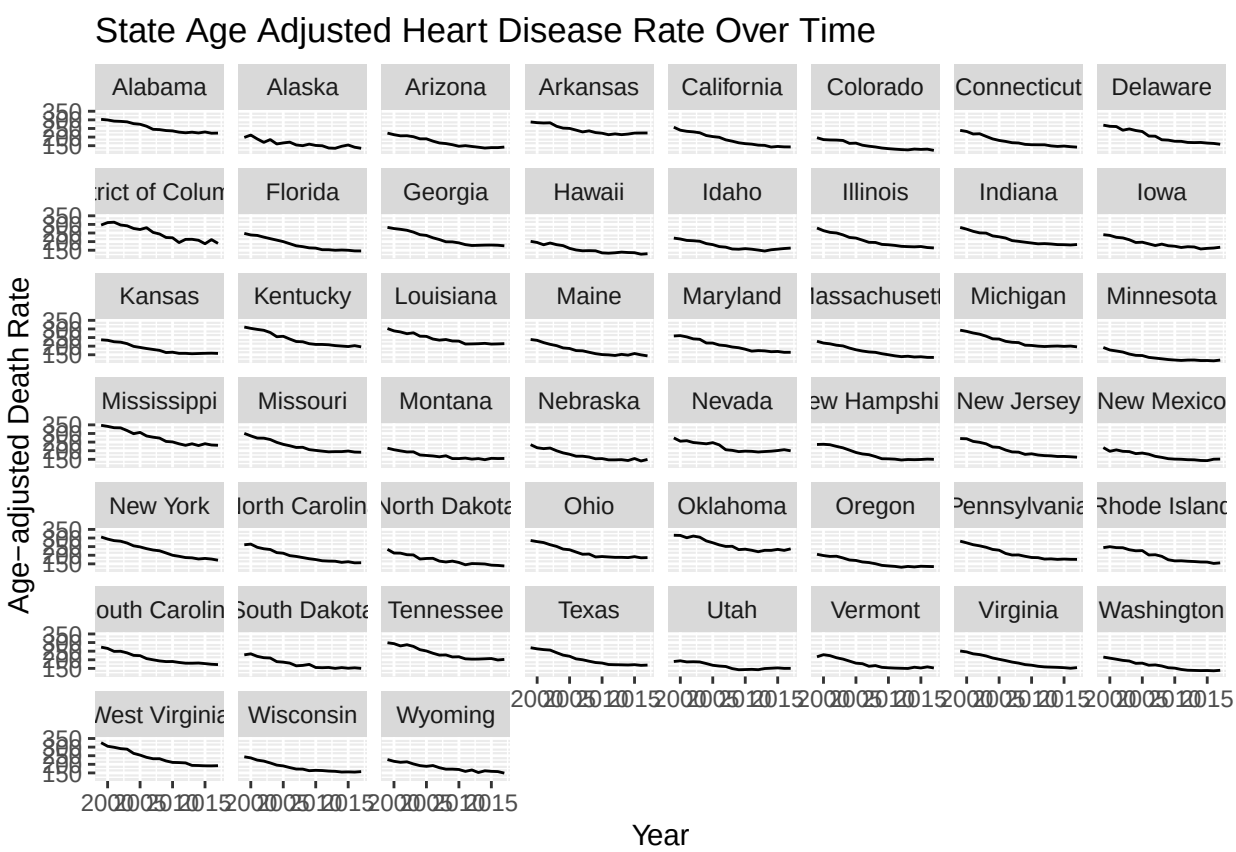



```r
# state level

state_only <- filtered_data |>
  filter(State != "United States") |>
  mutate(
    Region = case_when(
      State %in% c("Maine", "New Hampshire", "Vermont", "Massachusetts",
                   "Rhode Island", "Connecticut", "New York", "New Jersey",
                   "Pennsylvania") ~ "Northeast",
      State %in% c("Ohio", "Indiana", "Illinois", "Michigan", "Wisconsin",
                   "Minnesota", "Iowa", "Missouri", "North Dakota", "South Dakota",
                   "Nebraska", "Kansas") ~ "Midwest",
      State %in% c("Delaware", "Maryland", "District of Columbia", "Virginia",
```

```
                    "West Virginia", "North Carolina", "South Carolina", "Georgia",
                    "Florida", "Kentucky", "Tennessee", "Alabama", "Mississippi",
                    "Arkansas", "Louisiana", "Oklahoma", "Texas") ~ "South",
      State %in% c("Montana", "Idaho", "Wyoming", "Colorado", "New Mexico",
                    "Arizona", "Utah", "Nevada", "Washington", "Oregon", "California",
                    "Alaska", "Hawaii") ~ "West"
    )
  )


head(state_only)
```

```
## # A tibble: 6 x 7
##    Year `113 Cause Name` `Cause Name` State Deaths Age-adjusted Death R~1 Region
##   <dbl> <chr>            <chr>        <chr>  <dbl>                   <dbl> <chr>
## 1  2017 Diseases of hea~ Heart disea~ Alab~  13110                    223. South
## 2  2017 Diseases of hea~ Heart disea~ Alas~    814                    135  West
## 3  2017 Diseases of hea~ Heart disea~ Ariz~  12398                    142. West
## 4  2017 Diseases of hea~ Heart disea~ Arka~   8270                    224. South
## 5  2017 Diseases of hea~ Heart disea~ Cali~  62797                    143. West
## 6  2017 Diseases of hea~ Heart disea~ Colo~   7060                    123. West
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

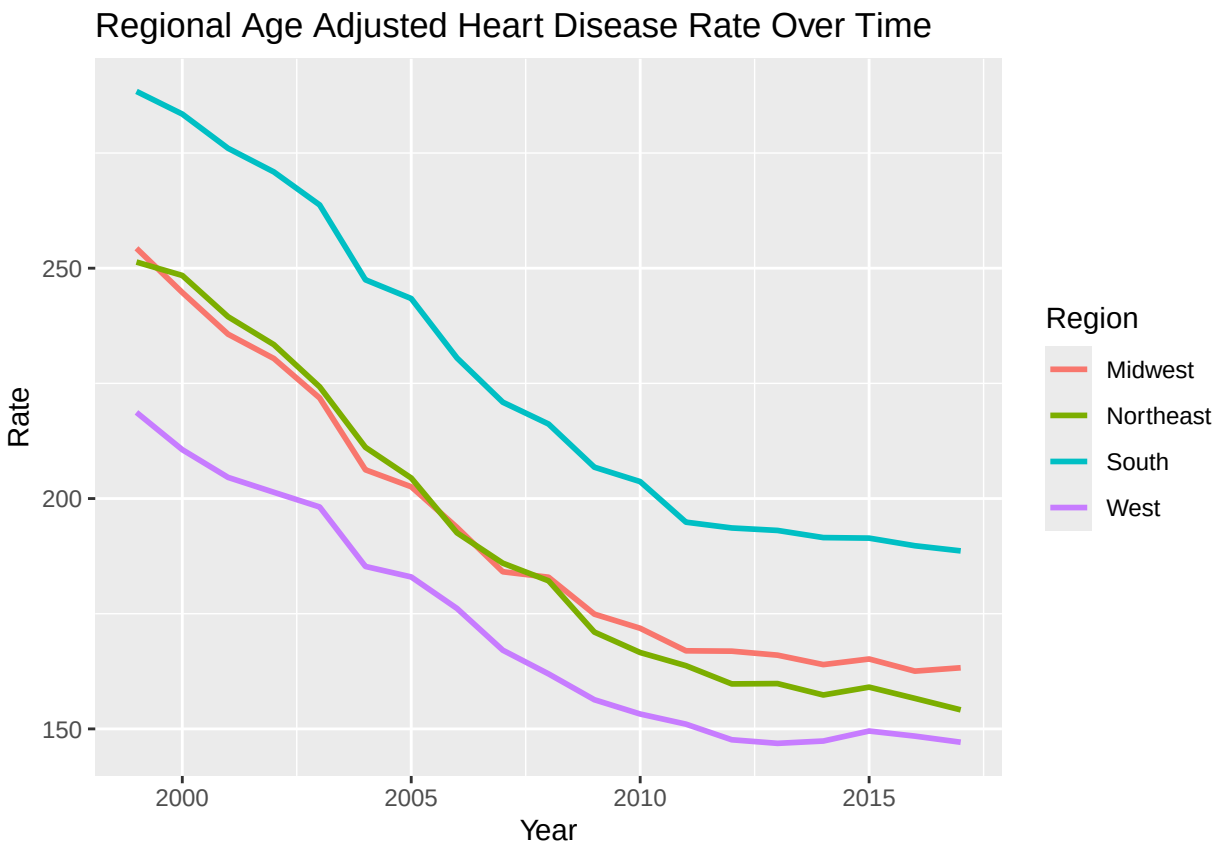
```
ggplot(data = state_only, mapping = aes(x = Year, y = `Age-adjusted Death Rate`)) +
  geom_line() +
  facet_wrap(~ State) +
  labs(title = "State Age Adjusted Heart Disease Rate Over Time")
```



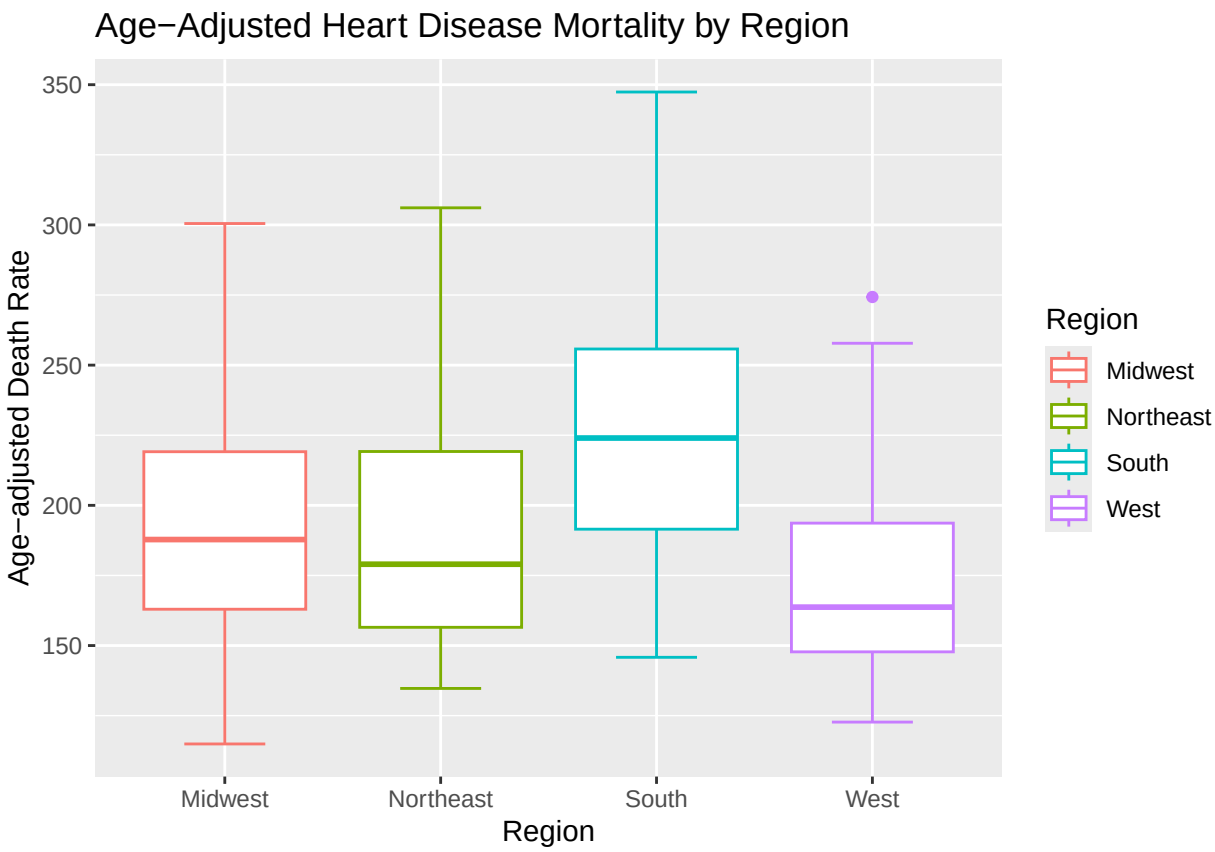
```r
# regional level
regional_summary <- state_only |>
  group_by(Region, Year) |>
  summarise(
    Rate = mean(`Age-adjusted Death Rate`)
  )
```

```
## `summarise()` has grouped output by 'Region'. You can override using the
## `.groups` argument.
```

```r
ggplot(data = regional_summary, mapping = aes(x = Year, y = Rate, color = Region)) +
  geom_line(linewidth = 1) +
  labs(title = "Regional Age Adjusted Heart Disease Rate Over Time")
```




```r
ggplot(data = state_only, mapping = aes(x = Region, y = `Age-adjusted Death Rate`, color = Region)) +
  geom_boxplot(staplewidth = 0.5) +
  labs(title = "Age-Adjusted Heart Disease Mortality by Region")
```



Statistical Testing

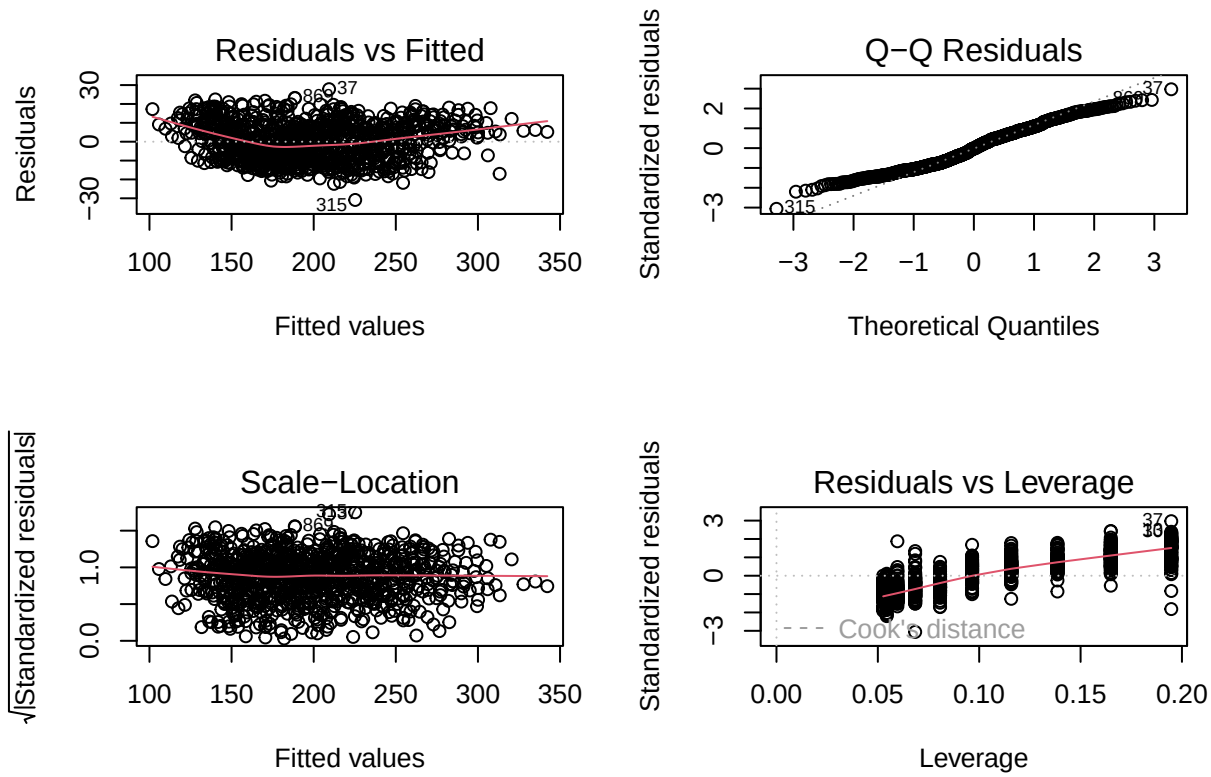
- How does mortality vary across states, and are there regional/geographical patterns over time?
- one way anova limitations:
  - has to be done using a single year, not all years
  - anova assumes independent groups
  - using all years violate independence
- two way anova
- notes:
  - https://www.geeksforgeeks.org/data-science/analysis-of-variance-anova/
  - cook's distance
    - How does a single observation (data pt) influece the entire model?
    - If a data pt is removed, does the model's prediction change a lot?

```
# two way anova testing
# * (interaction term) bc each state has their own slope, if + was used then slope for Year is the same
two_way <- aov(`Age-adjusted Death Rate` ~ State * Year, data = state_only)
summary(two_way)
```

```
##              Df Sum Sq Mean Sq  F value Pr(>F)
## State        50 957770   19155  175.520 <2e-16 ***
## Year          1 842449  842449 7719.314 <2e-16 ***
## State:Year   50  37132     743    6.805 <2e-16 ***
## Residuals   867  94620     109
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# check assumptions: homoscedasticity, normalit of residuals, linearity, and influential pts
par(mfrow = c(2,2))
plot(two_way)
```




```r
D <- cooks.distance(two_way)
sum(D > 1)
```

```
## [1] 0
```

Predictive Modeling

- Can past trends in age adjusted heart disease death rates be used to predict future state level rates?
- notes:
  - simple linear regression wont work bc we need year and state
    - assumes all states share the same mean and trend over time
  - multiple linear regression year + state
    - treats state as a categorical predictor (fixed state effects)
    - each state getes it's own intercept (baseline level)
    - BUT it assumes same slope for all states
  - multiple linear regression year * state
    - different slope per state

```r
# simple linear regression
m1 <- lm(`Age-adjusted Death Rate` ~ Year, data = state_only)
#summary(m1)
```

```r
# multiple regression +
m2 <- m_additive <- lm(`Age-adjusted Death Rate` ~ Year + State, data = state_only)
#summary(m2)

# multiple regression *
m3 <- lm(`Age-adjusted Death Rate` ~ Year * State, data = state_only)
#summary(m3)

# mixed model?

mixed1 <- lmer(`Age-adjusted Death Rate` ~ Year + (1 | State), data = state_only)
#summary(mixed1)

mixed2 <- lmer(`Age-adjusted Death Rate` ~ Year + (Year | State), data = state_only)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## unable to evaluate scaled gradient
```

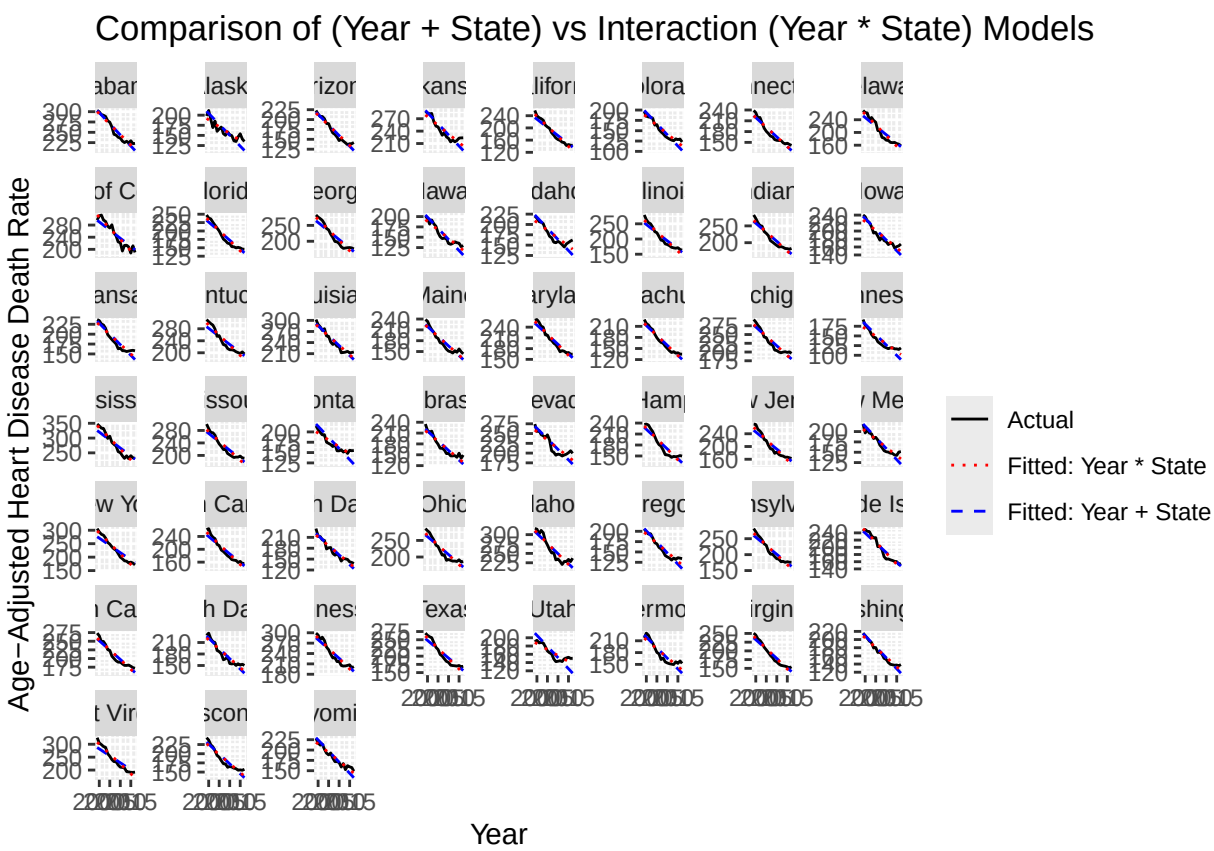
```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge: degenerate Hessian with 2 negative eigenvalues
```

```r
#summary(mixed2)


# plot comparison

state_only <- state_only |>
  mutate(
    fitted_m1 = fitted(m1),
    fitted_m2 = fitted(m2),
    fitted_m3 = fitted(m3),
    fitted_mixed1 = fitted(mixed1),
    fitted_mixed2 = fitted(mixed2)
  )

head(state_only)
```

```
## # A tibble: 6 x 12
##    Year `113 Cause Name` `Cause Name` State Deaths Age-adjusted Death R~1 Region
##   <dbl> <chr>            <chr>        <chr>  <dbl>                  <dbl> <chr>
## 1  2017 Diseases of hea~ Heart disea~ Alab~  13110                   223. South
## 2  2017 Diseases of hea~ Heart disea~ Alas~    814                   135  West
## 3  2017 Diseases of hea~ Heart disea~ Ariz~  12398                   142. West
## 4  2017 Diseases of hea~ Heart disea~ Arka~   8270                   224. South
## 5  2017 Diseases of hea~ Heart disea~ Cali~  62797                   143. West
## 6  2017 Diseases of hea~ Heart disea~ Colo~   7060                   123. West
## # i abbreviated name: 1: `Age-adjusted Death Rate`
## # i 5 more variables: fitted_m1 <dbl>, fitted_m2 <dbl>, fitted_m3 <dbl>,
## #   fitted_mixed1 <dbl>, fitted_mixed2 <dbl>
```

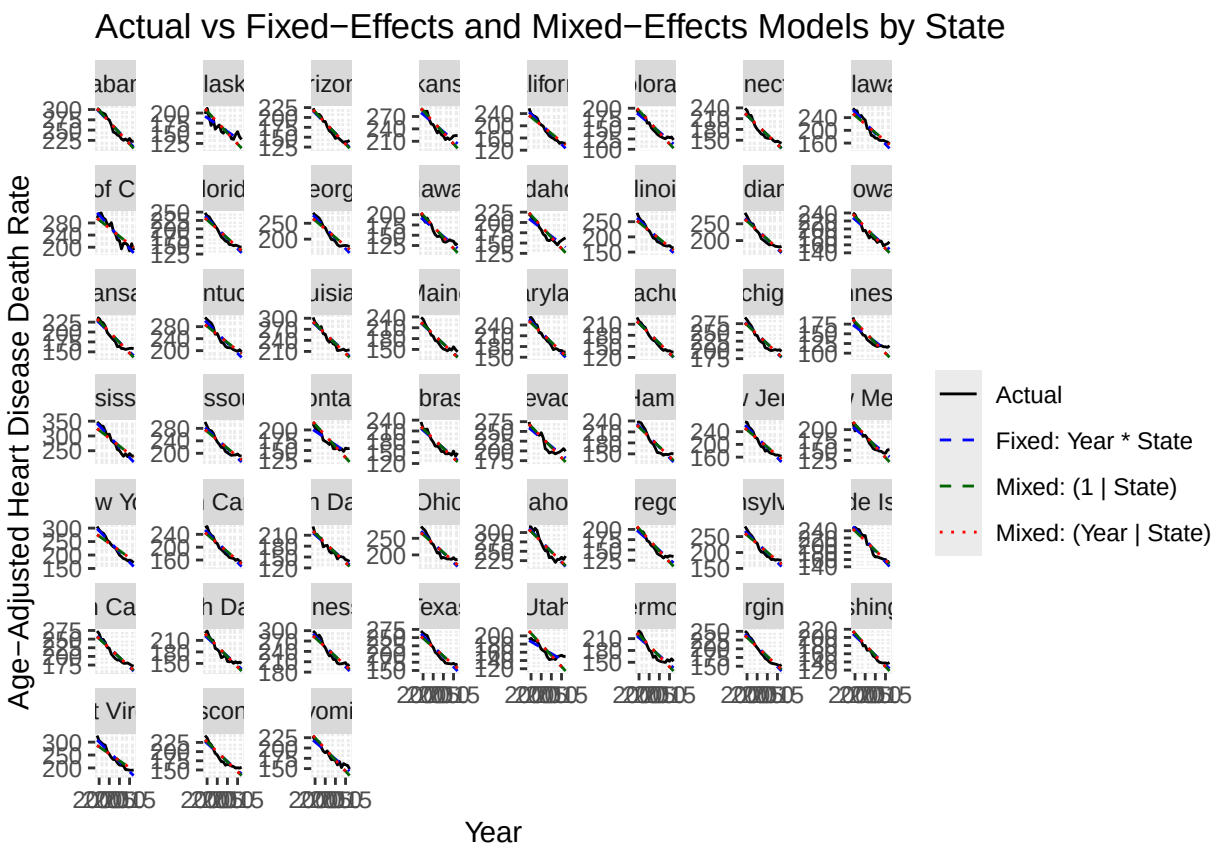
```r
ggplot(data = state_only, mapping = aes(x = Year)) +

  # actual data
  geom_line(mapping = aes(y = `Age-adjusted Death Rate`, color = "Actual")) +
```

```r
  # year + state
  geom_line(mapping = aes(y = fitted_m2, color = "Fitted: Year + State"), linetype = "dashed") +

  # year * state
  geom_line(mapping = aes(y = fitted_m3, color = "Fitted: Year * State"), linetype = "dotted") +

  facet_wrap(~ State, scales = "free_y") +

  labs(
    title = "Comparison of (Year + State) vs Interaction (Year * State) Models",
    y = "Age-Adjusted Heart Disease Death Rate",
    color = ""
  ) +

  scale_color_manual(values = c(
    "Actual" = "black",
    "Fitted: Year + State" = "blue",
    "Fitted: Year * State" = "red"
  ))
```




```r
ggplot(data = state_only, mapping = aes(x = Year)) +
  # actual data
  geom_line(mapping = aes(y = `Age-adjusted Death Rate`, color = "Actual")) +

  # m3
  geom_line(aes(y = fitted_m3, color = "Fixed: Year * State"), linetype = "dashed") +
```

```r
  # mixed1
  geom_line(aes(y = fitted_mixed1, color = "Mixed: (1 | State)"), linetype = "dashed") +

  # mixed2
  geom_line(mapping = aes(y = fitted_mixed2, color = "Mixed: (Year | State)"), linetype = "dotted") +

  facet_wrap(~ State, scales = "free_y") +

  labs(
    title = "Actual vs Fixed-Effects and Mixed-Effects Models by State",
    y = "Age-Adjusted Heart Disease Death Rate",
    color = ""
  ) +

  scale_color_manual(values = c(
    "Actual"               = "black",
    "Fixed: Year * State" = "blue",
    "Mixed: (1 | State)"  = "darkgreen",
    "Mixed: (Year | State)" = "red"
  ))
```



- Train Models:
  - 80/10/10 does not work bc we cant randomly shuffle years

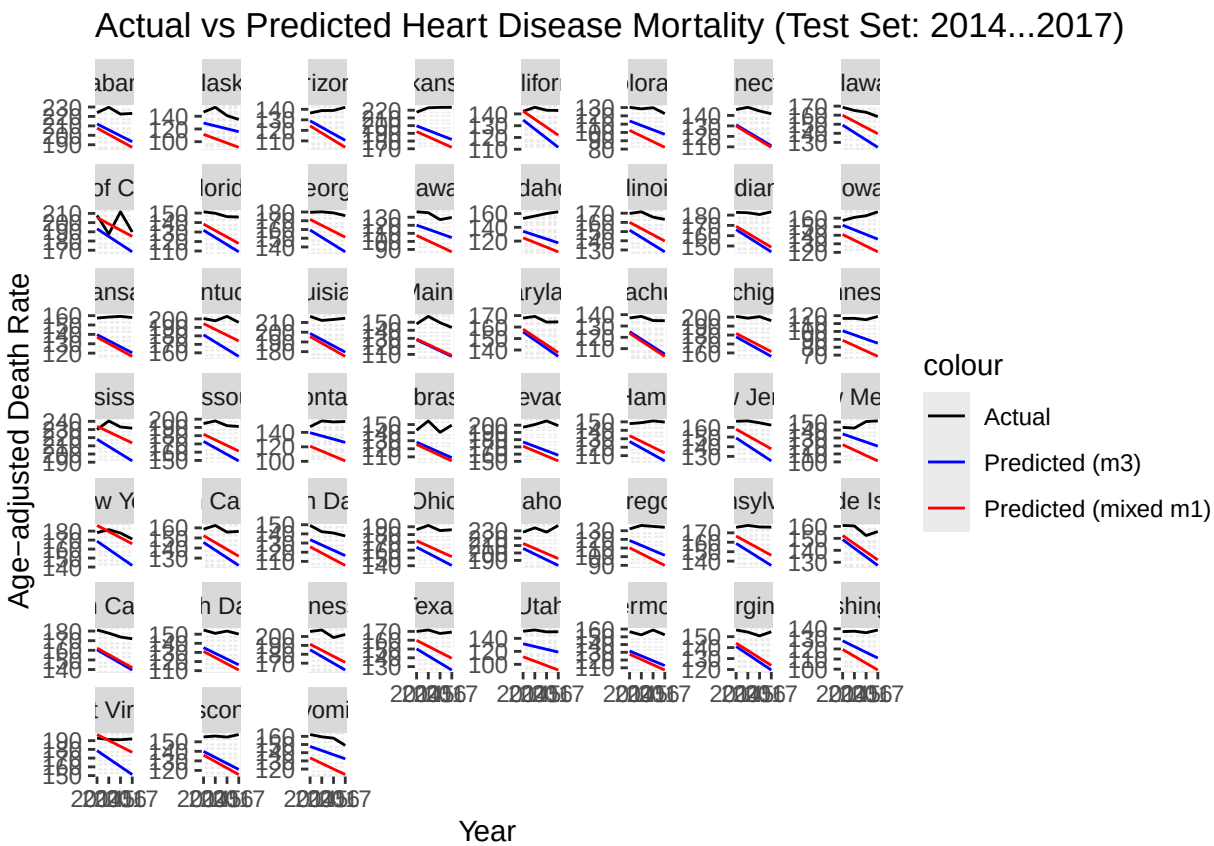
```r
train <- state_only |>
  filter(Year <= 2013)

test <- state_only |>
```

```r
  filter(Year > 2013)

# train models
m2_train <- lm(`Age-adjusted Death Rate` ~ Year + State, data = train)
m3_train <- lm(`Age-adjusted Death Rate` ~ Year * State, data = train)
mixed1_train <- lmer(`Age-adjusted Death Rate` ~ Year + (1 | State), data = train)
mixed2_train <- lmer(`Age-adjusted Death Rate` ~ Year + (Year | State), data = train)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
# predict on test set
test$pred_m2 <- predict(m2_train, newdata = test)
test$pred_m3 <- predict(m3_train, newdata = test)
test$pred_mixed1 <- predict(mixed1_train, newdata = test)
test$pred_mixed2 <- predict(mixed2_train, newdata = test)

ggplot(data = test, mapping = aes(x = Year)) +

  geom_line(mapping = aes(y = `Age-adjusted Death Rate`, color = "Actual")) +

  geom_line(mapping = aes(y = pred_m3, color = "Predicted (m3)")) +

  geom_line(mapping = aes(y = pred_mixed1, color = "Predicted (mixed m1)")) +

  facet_wrap(~ State, scales = "free_y") +

  labs(title = "Actual vs Predicted Heart Disease Mortality (Test Set: 2014-2017)") +

  scale_color_manual(values = c("Actual" = "black", "Predicted (m3)" = "blue", "Predicted (mixed m1)" =
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <93>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <80>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality (Test Set:
## 2014-2017)' in 'mbcsToSbcs': dot substituted for <93>
```



Evaluation and Interpretation

- look at RMSE and MAE

```
# check models accuracies
# RMSE and MAE
rmse_m2 <- rmse(test$`Age-adjusted Death Rate`, test$pred_m2)
mae_m2  <- mae(test$`Age-adjusted Death Rate`, test$pred_m2)

rmse_m3 <- rmse(test$`Age-adjusted Death Rate`, test$pred_m3)
mae_m3  <- mae(test$`Age-adjusted Death Rate`, test$pred_m3)

rmse_mixed1 <- rmse(test$`Age-adjusted Death Rate`, test$pred_mixed1)
mae_mixed1  <- mae(test$`Age-adjusted Death Rate`, test$pred_mixed1)

rmse_mixed2 <- rmse(test$`Age-adjusted Death Rate`, test$pred_mixed2)
mae_mixed2  <- mae(test$`Age-adjusted Death Rate`, test$pred_mixed2)

# accuracies
data.frame(
  Model = c("m2 (Year+State)", "m3 (Year*State)",
            "mixed1 (1|State)", "mixed2 (Year|State)"),
  RMSE = c(rmse_m2, rmse_m3, rmse_mixed1, rmse_mixed2),
  MAE  = c(mae_m2, mae_m3, mae_mixed1, mae_mixed2)
)
```

```
##                   Model      RMSE       MAE
```

```
## 1     m2 (Year+State) 28.99761 26.20274
## 2     m3 (Year*State) 27.51421 25.92697
## 3    mixed1 (1|State) 28.96682 26.19361
## 4 mixed2 (Year|State) 28.98851 26.19928
```

Between the two models, the predictions follow a smooth linear decline because the models learn long term state specific trends but it can't capture short term year to year flucatuatons. Based on the RMSE and MAE Year x State (m3) performed the best as it showed the lowest scores, whicbh means allowing each state to have its own slope improves predictive model performance. Mixed effects were a bit less accurate because it tries to make the slopes to be more similar across states. This means that past linear trends alone are not enough for accurate year to year predictions of state level heart disease mortality.

Random Forest

```r
library(randomForest)
set.seed(54)
train$State = factor(train$State)
test$State = factor(test$State)

rm_model = randomForest(`Age-adjusted Death Rate`~Year+State, data = train, ntree = 1000, mtry = 2, impo

print(rm_model)
```

```
##
## Call:
##  randomForest(formula = `Age-adjusted Death Rate` ~ Year + State,      data = train, ntree = 1000, mt
##                Type of random forest: regression
##                      Number of trees: 1000
## No. of variables tried at each split: 2
##
##           Mean of squared residuals: 38.46033
##                     % Var explained: 98.08
```

```r
test$pred_rm = predict(rm_model, newdata=test)
rmse_rfm = rmse(test$`Age-adjusted Death Rate`, test$pred_rm)
mae_rfm = mae(test$`Age-adjusted Death Rate`, test$pred_rm)
```

```r
data.frame(
  Model = c("m2 (Year+State)", "m3 (Year*State)",
            "mixed1 (1|State)", "mixed2 (Year|State)", "RF (Year + State"),
  RMSE = c(rmse_m2, rmse_m3, rmse_mixed1, rmse_mixed2, rmse_rfm),
  MAE  = c(mae_m2, mae_m3, mae_mixed1, mae_mixed2, mae_rfm)
)
```

```
##                 Model     RMSE       MAE
## 1     m2 (Year+State) 28.99761 26.202744
## 2     m3 (Year*State) 27.51421 25.926974
## 3    mixed1 (1|State) 28.96682 26.193606
## 4 mixed2 (Year|State) 28.98851 26.199278
## 5    RF (Year + State  6.29923  5.015603
```

```r
# R^2
y_test = test$`Age-adjusted Death Rate`
y_rmf = test$pred_rm

ss_res = sum((y_test - y_rmf)^2)
ss_tot = sum((y_test-mean(y_test))^2)
```

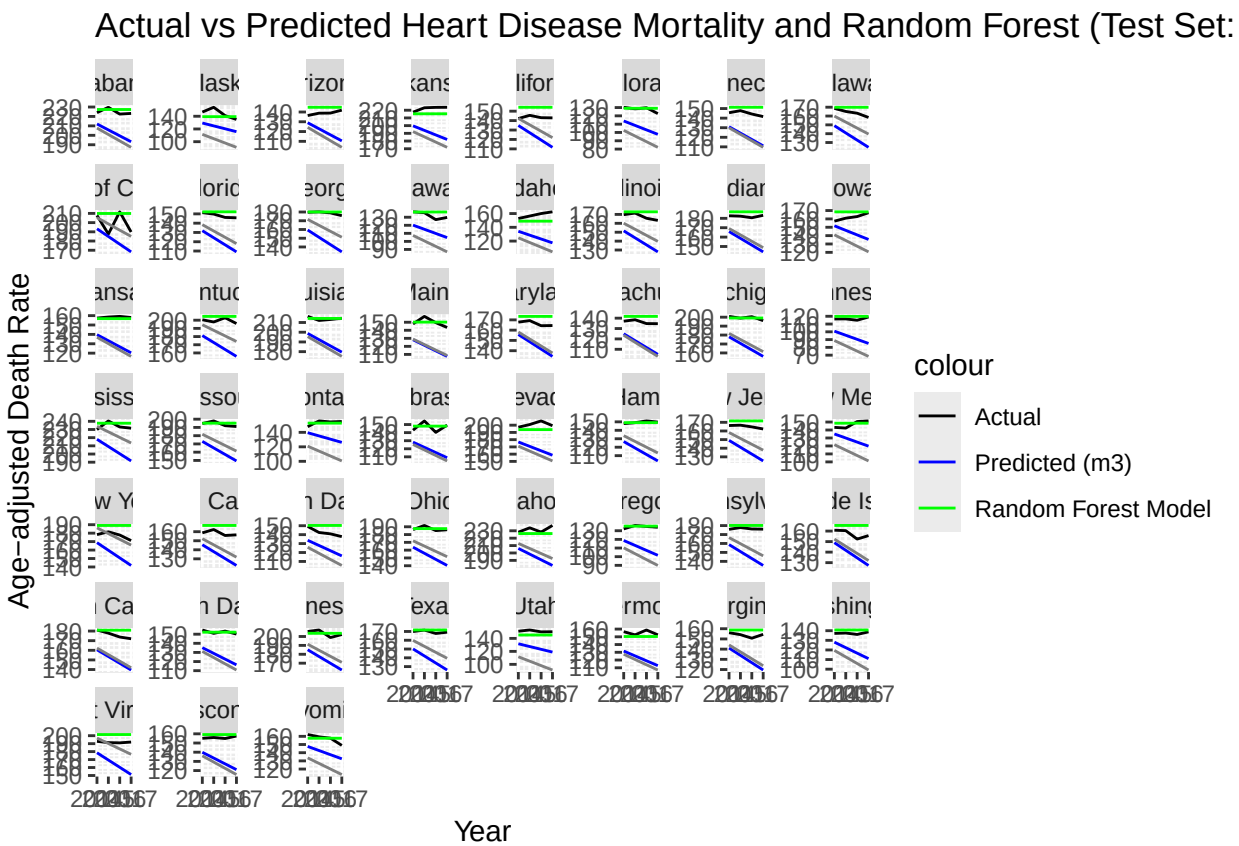
```r
R2_rmf = 1 - ss_res/ss_tot

R2_rmf
```

```
## [1] 0.9497093
```

So about 95% of the variation in state-level age-adjusted heart-disease mortality(2014-2017) is explained by the random forest model using only Year and State

```r
ggplot(data = test, mapping = aes(x = Year)) +

  geom_line(mapping = aes(y = `Age-adjusted Death Rate`, color = "Actual")) +

  geom_line(mapping = aes(y = pred_m3, color = "Predicted (m3)")) +

  geom_line(mapping = aes(y = pred_mixed1, color = "Predicted (m1)")) +

  geom_line(mapping = aes(y = pred_rm, color = "Random Forest Model"))+

  facet_wrap(~ State, scales = "free_y") +

  labs(title = "Actual vs Predicted Heart Disease Mortality and Random Forest (Test Set: 2014-2017)") +

  scale_color_manual(values = c("Actual" = "black", "Predicted (m3)" = "blue", "Predicted (mixed m1)" =
```

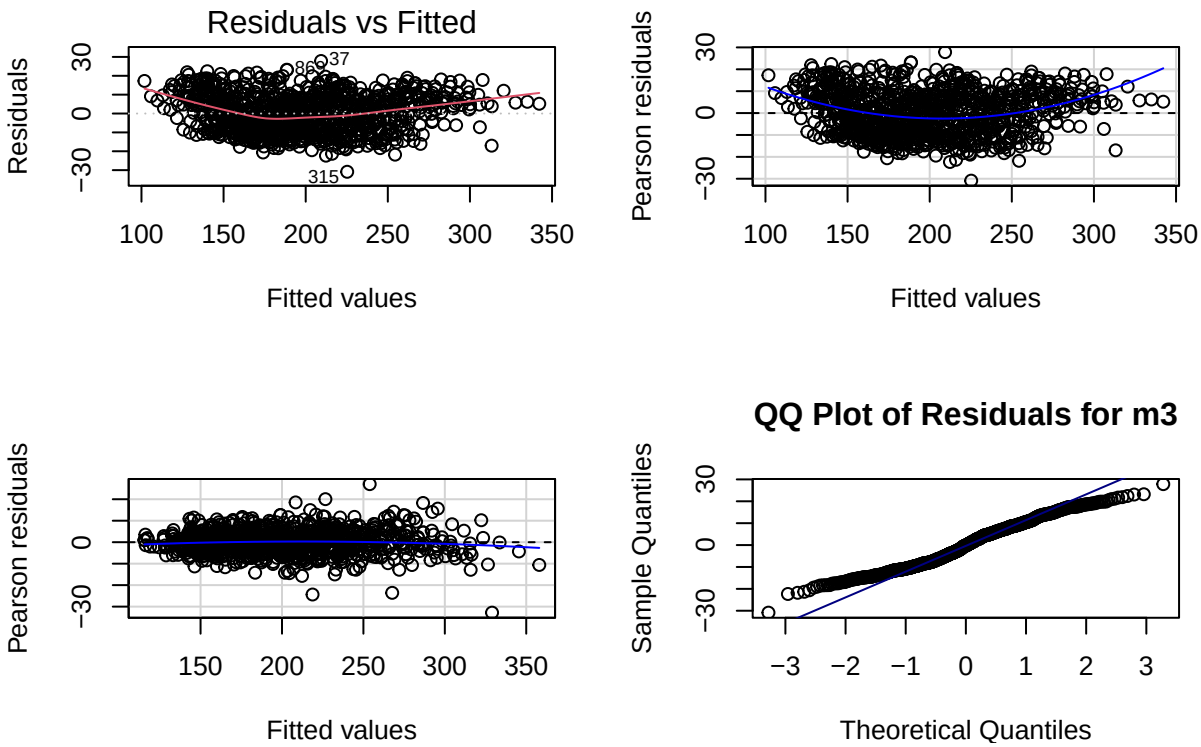
```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality and Random
## Forest (Test Set: 2014-2017)' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality and Random
## Forest (Test Set: 2014-2017)' in 'mbcsToSbcs': dot substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality and Random
## Forest (Test Set: 2014-2017)' in 'mbcsToSbcs': dot substituted for <93>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality and Random
## Forest (Test Set: 2014-2017)' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality and Random
## Forest (Test Set: 2014-2017)' in 'mbcsToSbcs': dot substituted for <80>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Actual vs Predicted Heart Disease Mortality and Random
## Forest (Test Set: 2014-2017)' in 'mbcsToSbcs': dot substituted for <93>
```




```
#Chapter 8 Diagnostic Tests

#Residulas vs Fitted Values
m3 <- lm(`Age-adjusted Death Rate` ~ State * Year, data = state_only)
par(mfrow=c(2,2))
plot(m3, 1)

#Curvature Test
residualPlot(m3) # gives quadratic test for year

#Additional Test - add a quadratic year term
m3_quad = update(m3,~.+I(Year^2))

#summary(m3)
#summary(m3_quad)

residualPlot(m3_quad)

#Cook Test
D_m3 <- cooks.distance(m3)
sum(D_m3 > 1)
```

```
## [1] 0
```

```
qqnorm(resid(m3), main = "QQ Plot of Residuals for m3")
qqline(resid(m3), col = "navy")
```

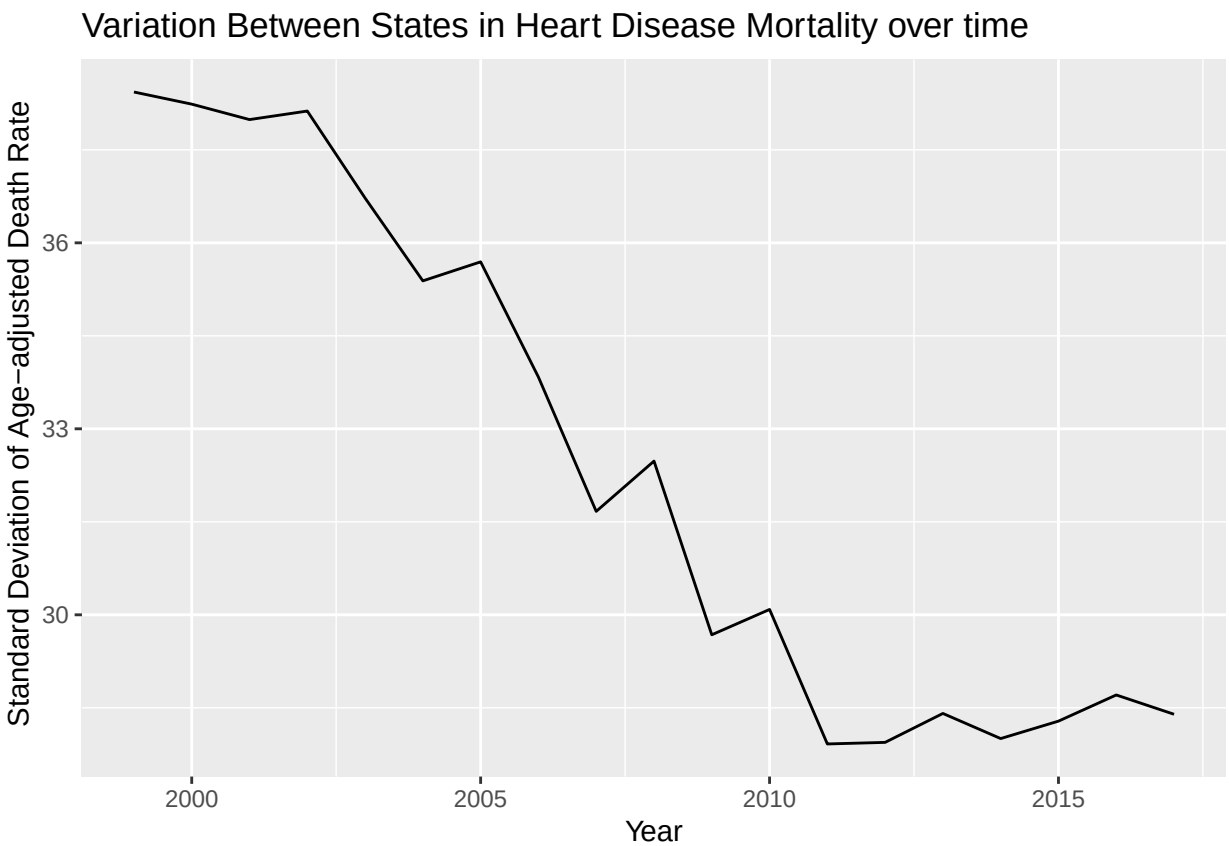



```r
#Extra - Shapiro-Wilk Test
shapiro.test(resid(m3))
```

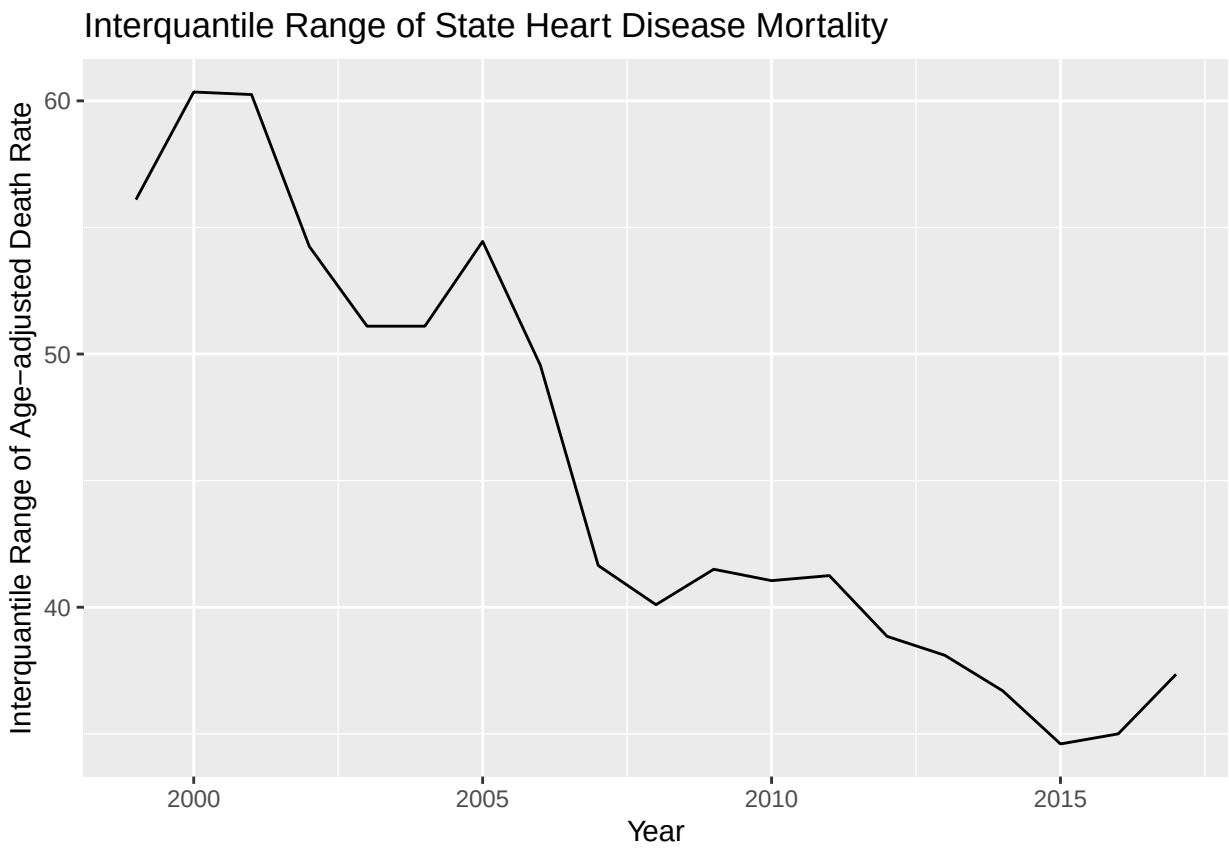
```
##
##  Shapiro-Wilk normality test
##
## data:  resid(m3)
## W = 0.97963, p-value = 2.142e-10
```

#Quantinfying whether states are becoming more similar or different

```r
state_range = state_only |>
  group_by(Year) |>
  summarise(mean_val = mean(`Age-adjusted Death Rate`),
            sd_val = sd(`Age-adjusted Death Rate`),
            iqr_val = IQR(`Age-adjusted Death Rate`))

ggplot(state_range, aes(x = Year, y = sd_val))+
  geom_line()+
  labs(
    title = "Variation Between States in Heart Disease Mortality over time",
    y = " Standard Deviation of Age-adjusted Death Rate"
  )
```




```r
ggplot(state_range, aes(x = Year, y = iqr_val))+
  geom_line()+
  labs(
    title = "Interquantile Range of State Heart Disease Mortality",
    y = "Interquantile Range of Age-adjusted Death Rate"
  )
```



Both SD and IQR of state heart-disease mortality fell While overall rates declined nationally, state rates became more similar to each other